

Photovoltaic Knowledge-Informed Neural Network (PKINN): Interpretable power prediction model under Fluctuating Environmental Conditions

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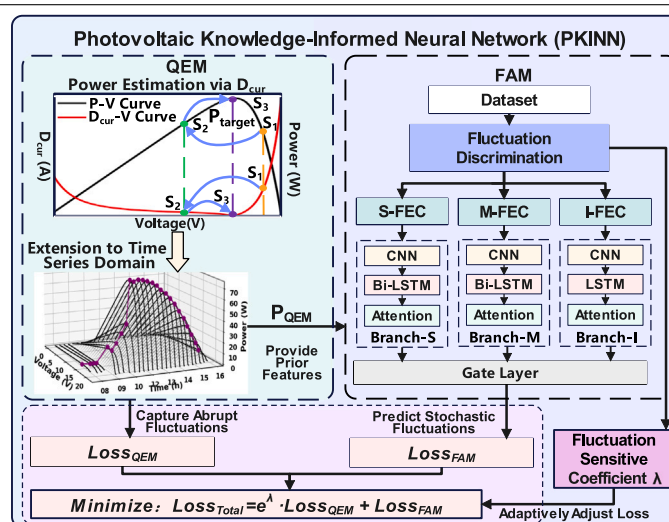
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HIGHLIGHTS

- PKINN adaptively learns fluctuation patterns under diverse environmental conditions.
- QEM formulates explicit expressions to capture abrupt PV power fluctuations.
- FAM quantifies stochastic fluctuations across conditions with different intensities.
- Experiments demonstrate that PKINN reduces the RMSE of predictions by over 8.73%.

GRAPHICAL ABSTRACT



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ABSTRACT

Fluctuating Environmental Conditions (FECs) are a critical barrier to accurate photovoltaic (PV) power forecasting. Existing models often fail to capture abrupt and stochastic fluctuations, leading to reduced forecasting reliability. To address this challenge, this study proposes an interpretable Photovoltaic Knowledge-Informed Neural Network (PKINN). The framework incorporates a Quadratic Explicit Model (QEM) to derive explicit expressions of PV power and transparently capture abrupt variations, while a Fluctuation Allocation Mechanism (FAM) employs a fluctuation sensitivity coefficient to quantify fluctuation intensity and allocate input data to specialized prediction branches. The proposed PKINN framework enables adaptive learning across diverse FECs and enhances forecasting performance. Experimental evaluations on two types of PV modules demonstrate that PKINN reduces the root mean square error by at least 8.73% compared with state-of-the-art models across diverse FECs.

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1. Introduction

Growing energy shortages and environmental degradation caused by fossil fuel combustion have intensified the demand for renewable energy [1]. Solar energy, with its wide availability and pollution-free characteristics, holds significant potential for development [2]. Photovoltaic (PV) systems are typically deployed as distributed power sources to ensure local electricity supply, which is crucial for optimizing energy management and resource allocation. By predicting PV power output, power systems can optimize charging strategies and prevent grid frequency instability and voltage anomalies [3]. Therefore, accurate PV power prediction is essential for ensuring the safety, economic efficiency, and stability of power systems [4]. However, unpredictable fluctuations in environmental variables such as irradiance and temperature frequently destabilize PV system performance. Such non-stationary conditions, resulting from natural environmental variability, are referred to as Fluctuating Environmental Conditions (FECs) [5]. FECs are primarily caused by several factors. Firstly, sudden meteorological events such as heavy rainfall, thunderstorms, and cumulus clouds often accompany rapid environmental changes, leading to abrupt fluctuations in PV power output [6]. Secondly, haze, dust, and suspended particles in the atmosphere can induce persistent random disturbances in environmental variables, gradually accumulating into fluctuation noise [7]. Additionally, localized variations in irradiance caused by the shading of buildings or trees can also affect the operating state of PV modules. The abrupt power fluctuations induced by FECs make it difficult for prediction models to respond promptly to environmental changes, while noise disturbances further exacerbate the nonlinearity and uncertainty between input features and power output [8]. Consequently, PV power prediction faces compounded challenges under FECs arising from abrupt ramp events and stochastic fluctuations, which pose demanding demands on temporal responsiveness and uncertainty modeling capability.

PV power prediction models are generally classified into physical models, data-driven models, and ensemble models [9]. Physical models construct equivalent PV cell models using PV module parameters and predict power output based on physical principles. They can capture power variation trends across different conditions and maintain reliability even during abrupt environmental changes [10]. A category of methods employs simplified physical models that approximate complex physical processes. For example, Salah et al. [11] proposed an Empirical Approximation Model (EAM) that estimates parameters to derive functions for irradiance and temperature, enabling peak power prediction under various operating conditions. In [12], a model utilizing a steady-state thermal method to simulate PV module temperature was proposed, which estimates power output using calibrated characteristic parameters of PV panels. Additionally, some methods consider the strong influence of environmental factors on PV power output, incorporating dynamic environmental parameters into physical models to refine prediction accuracy. For instance, Lopes et al. [13] developed a general heuristic physics-based prediction model based on monthly irradiance, which used chain analysis to select the appropriate solar model for specific regions. To improve the reliability of environmental data in predictions, Mayer et al. [14] integrated a weather prediction model with model chains to predict irradiance and applied quantile regression calibration for PV power forecasting. In [15], a rooftop PV power forecasting physical model that combined environmental parameters with an improved power tracking algorithm was proposed to classify and predict power variation patterns. Physical models can capture power trends driven by environmental factors under FECs and reflect abrupt changes. However, their reliance on idealized physical assumptions restricts their ability to represent long-term degradation effects such as module aging and soiling. These models lack effective interaction with environmental data, which limits their capacity to fit stochastic fluctuations driven by complex environmental disturbances.

Data-driven models learn nonlinear and spatiotemporal dependencies from data, enabling them to capture uncertainties in environmental conditions and mitigate the influence of noise and random disturbances on prediction [16,17]. They generally include statistical models, machine-learning models, and deep-learning models. Statistical models rely on distributional assumptions and probabilistic analysis, and are mostly applied to short-term or near-linear scenarios. The Autoregressive Moving Average (ARMA), Autoregressive Integrated Moving Average (ARIMA), and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models have been widely adopted in PV power forecasting. Lauret et al. [18] combined ARMA, which captures the linear dependency structure of time series, with GARCH, which models conditional heteroskedasticity, to achieve probabilistic forecasts of irradiance and PV power. In [19], ARIMA was applied to extract trend and periodic components, thereby improving short-term forecasting accuracy. These approaches, however, are constrained by their linear assumptions and thus fail to represent the nonlinear interactions and complex dynamics inherent in PV power generation. Machine learning models employ adaptive algorithms to extract features from PV data. For instance, Pan et al. [20] used Support Vector Machines (SVM) to capture nonlinear dependencies between PV power and environmental variables, improving accuracy via enhanced parameter optimization. Chen et al. [21] integrated chaotic property analysis and Particle Swarm Optimization (PSO) with the Kernel Extreme Learning Machine (KELM) [22] to achieve online forecasting. In [23], a hybrid approach combined multi-scale similar day analysis with ESN-KELM, where an improved Archimedes algorithm was employed for parameter optimization and ensemble aggregation. Nevertheless, the performance of such models heavily depends on feature engineering and parameter tuning, limiting robustness and generalization across diverse operational conditions. Deep-learning models employ deep network architectures to capture hierarchical and multi-scale data representations, enabling more comprehensive modeling of complex dependencies. For example, Luo et al. [24] performed sensitivity analysis for input feature selection and applied Long Short-Term Memory (LSTM) networks to predict hourly PV power. In [25], a day-ahead model integrated a General Regression Neural Network (GRNN) with the Grey Wolf Optimizer (GWO) for parameter learning. Yuan et al. [26] improved the Transformer by introducing an MFA-attention mechanism, allowing the model to capture fluctuations across multiple time scales and enhancing responsiveness to local variations. Furthermore, [27] presented an end-to-end Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) model capable of extracting periodic patterns and feature correlations in time series. Although deep-learning models have achieved notable success, their complex architectures and extensive parameterization restrict their adaptability to abrupt fluctuations. Overall, data-driven models suffer from inherent limitations in PV power forecasting that involve abrupt ramps, as they lack explicit modeling and corrective representation of the PV power generation process, which constrains the learning of ramp fluctuations and leads to delayed predictions or underestimated ramp magnitudes.

Ensemble models leverage complementary strengths of various models to address limitations in generalization capability and prediction stability of single-model approaches. Various ensemble models combining statistical and machine-learning models have been applied in PV power forecasting. These models integrate the robustness of statistical models with the nonlinear feature-extraction capability of machine-learning models [28]. For instance, Tan et al. [29] proposed a multi-source prediction model based on satellite cloud images, utilizing cloud motion trajectory data to provide environmental features for an LSTM model. Similarly, a proposed LSTM-ARIMA hybrid model [30] achieved more accurate power prediction by decomposing and reconstructing signals to segment different power intervals. In [31], Hong et al. introduced a digital twin system that integrates an LSTM with a triangular shadow pattern estimation method to predict power under shading

conditions. To address the insufficient extraction of spatiotemporal correlations in distributed PV power forecasting, Wang et al. [32] proposed an integrated two-dimensional distributed forecasting framework. This approach combines a self-attention mechanism, Bidirectional Convolutional Gated Recurrent Units (BiConvGRU), and an encoder-decoder architecture to capture the spatiotemporal evolution patterns of PV generation. Furthermore, previous works have employed ensemble models to address PV power ramp events, which are characterized by rapid and large-magnitude changes in power output over short time intervals. These models typically integrate data-driven forecasting models with sensitive mechanisms to improve robustness under highly unstable conditions. For instance, Song et al. [33] proposed a probabilistic ramp forecasting framework based on natural gradient boosting, capable of quantifying power uncertainty at sub-minute to minute scales. The ConvODE-Mixer model [34] combines convolutional encoding with dynamic system modeling to enhance its ability to capture rapid transient responses. Moreover, Abuella et al. [35] introduced a post-processing adjustment mechanism that refines ramp timing and magnitude estimation, thereby improving predictive accuracy under extreme fluctuations. Ensemble models enhance feature extraction and generalization through methodological complementarity compared with single-model approaches. However, they lack a unified characterization and regulation of fluctuation intensity variations, weakening their predictive performance for abrupt ramps and stochastic fluctuations, and leading to prediction instability under limited samples or distribution shifts.

Physics-Informed Neural Network (PINN) represents a class of deep learning methods that incorporate physical laws as constraints to guide the model toward solutions consistent with known physical principles [36]. PINN ensures that the model output adheres to known physical laws while accurately matching the observational data by enforcing physical information constraints within the loss function. In recent years, PINN has been widely applied in fields such as materials science, geophysics, and medicine [37,38], and has gradually expanded into the domain of energy systems. For instance, PINN has been used to predict three-dimensional spatiotemporal wind fields [39] and to model microgrid transients [40]. In these works, energy-related physical constraints are systematically embedded into machine learning frameworks, effectively mitigating the challenges posed by data scarcity and incomplete observations while enhancing model robustness and generalization. While PINN has demonstrated advantages in various domains, its application in PV power forecasting remains limited, as PV generation does not possess an explicit PDE formulation and cannot meet the prerequisites required for a PINN framework.

Based on the above review, the limitations of existing PV power forecasting methods can be summarized as follows:

- For abrupt power fluctuations induced by environmental changes, existing forecasting models often fail to promptly capture rapid variations in power output, leading to delayed responses or inaccurate estimation of power ramp magnitudes.
- For stochastic fluctuations, existing forecasting models lack effective mechanisms to characterize and differentiate fluctuation intensity across diverse FECs, resulting in insufficient robustness to noise disturbances.

These limitations indicate that existing forecasting methods lack systematic and explicit modeling of PV power fluctuation dynamics, particularly in capturing abrupt ramp behaviors and stochastic fluctuations under FECs. This gap highlights the need for a predictive framework that explicitly models power evolution while adapting to different fluctuation intensities. To this end, this paper proposes a Photovoltaic Knowledge-Informed Neural Network (PKINN). In the PKINN, the Quadratic Explicit Model (QEM) establishes analytical expressions to capture abrupt and ramp power fluctuations, while the Fluctuation Allocation Mechanism (FAM) quantifies fluctuation intensity to mitigate the impact of stochastic disturbances. The main contributions of this paper are as follows:

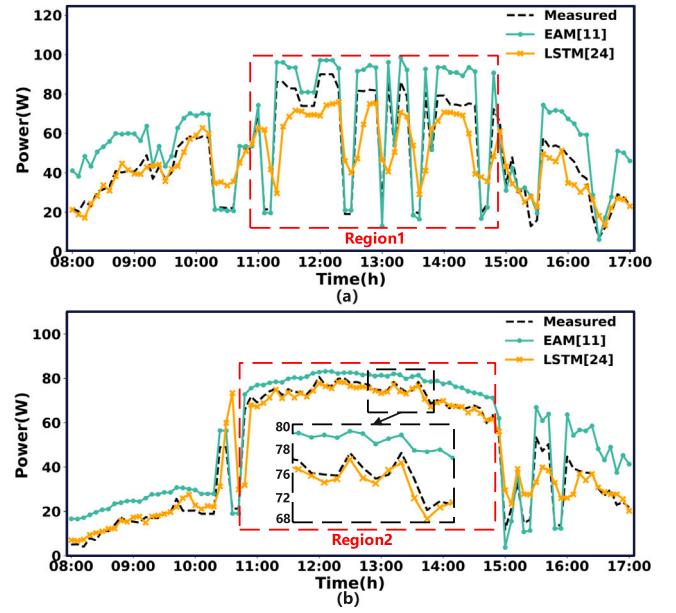


Fig. 1. Comparison of power prediction performance between the measured power and the predictions from EAM [11] and LSTM [24] under two types of scenarios: (a) abrupt power fluctuations under showery weather scenarios; (b) stochastic power fluctuations under foggy scenarios.

1. To address abrupt power fluctuations, this paper proposes a Quadratic Explicit Model (QEM). This model establishes explicit expressions of PV power to interpretably capture abrupt variations.
2. A Fluctuation Allocation Mechanism (FAM) is introduced to quantify stochastic fluctuations. Leveraging a fluctuation sensitivity coefficient, the mechanism guides the model to learn stochastic fluctuations under different fluctuation intensity conditions.
3. This paper develops an interpretable PKINN forecasting framework. The framework adaptively learns fluctuation features across diverse FECs and enhances prediction accuracy.

The remainder of this paper is organized as follows. Section 2 discusses the limitations of existing PV power forecasting methods. Section 3 provides a detailed description of the proposed PKINN. Section 4 presents the experimental performance and discussion of the prediction method. Finally, Section 5 concludes the paper.

2. Challenges in PV power prediction under FECs

In PV power forecasting tasks, the presence of FECs poses significant challenges. In particular, the abrupt fluctuations in power output, along with stochastic fluctuations stemming from environmental and system factors, compromise the predictive accuracy of existing models. This section systematically analyzes the characteristics of these fluctuations and discusses the specific challenges they impose on PV power prediction tasks.

One major type of FEC appears as abrupt fluctuations in PV power output, typically triggered by sudden weather events or transient shading conditions. The swift variations in irradiance or temperature induce a mismatch between environmental excitation and the PV module's thermo-electrical response. Due to this nonlinear and delayed adjustment process, the module cannot adjust rapidly, leading to pronounced short-term disturbances in power output. These fluctuations exhibit a high rate of change and considerable amplitude, disrupting the stationarity of power sequences and obscuring underlying trend patterns.

Consequently, they pose severe challenges to prediction models in maintaining temporal sensitivity and adaptive forecasting capacity. The adverse impact of such fluctuations is particularly pronounced in data-driven models that rely on the statistical distribution of historical samples. These models are typically trained by minimizing an objective function of the following form:

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|^2 \quad (1)$$

where $\hat{y}_i$ and y_i denote the predicted and observed power values, respectively, and N is the number of training samples [41]. This objective function measures the model's prediction accuracy by calculating the average error across the dataset. Under FECs, these abrupt fluctuations are highly diverse but insufficiently represented in training data. Due to the insufficient representation of various ramp patterns, data-driven models tend to generate overly smoothed predictions and lack sensitivity to abrupt power changes. During the forecasting process, such models often struggle to promptly identify input shifts induced by sudden environmental variations, leading to prediction bias, omission of fluctuations, and increased errors. For example, as shown in Fig. 1(a), Region 1 illustrates abrupt power fluctuations under showery weather scenarios. The LSTM [24], as a representative data-driven model, fails to accurately capture these fluctuations, with several predicted values exhibiting notable prediction shifts. This limitation mainly results from the smooth prediction strategies adopted during training, which reduce the model's sensitivity to actual power variations and cause it to overlook certain fluctuation patterns. In contrast, the physics-based Empirical Approximation Method (EAM) [11] accurately captures the timing and direction of fluctuations by explicitly modeling the intrinsic relationships between PV power and its influencing factors.

Stochastic fluctuations represent another major type of variation in PV power output. They arise from factors such as haze, dust deposition, and sensor errors, which are difficult to quantify and cannot be readily incorporated into explicit modeling frameworks. The randomness associated with such fluctuations often exhibits unstable spatiotemporal distributions and lacks clear causal relationships, resulting in unstructured and non-periodic power changes. These fluctuations pose significant challenges to the feature extraction and generalization capabilities of predictive models. Physics-based models are typically constructed from static or quasi-static electrical equations of PV modules. However, such models frequently fail to capture the stochasticity and nonlinear coupling introduced by disturbances in real-world applications. In addition, the absence of an interaction mechanism with complex fluctuation states limits their flexibility in adjusting architecture and parameters in response to changing conditions. As a result, a non-negligible discrepancy often arises between the theoretical outputs of physical models and the actual measured power values under noisy FECs. Fig. 1 illustrates the performance of the EAM as a representative physical model for PV power prediction. The EAM derives approximate computational formulas for PV modules by establishing a simulation circuit model of PV cells and applies a numerical iterative feedback method to estimate and predict PV power. Fig. 1(b) depicts a PV operating condition under foggy scenarios, where the fog is relatively dense in the morning, while Region 2 exhibits stochastic power fluctuations due to the diffused characteristics of the atmosphere. It is observed that the EAM consistently overestimates PV generation in Region 2 and fails to capture the subtle oscillations in power output. This limitation arises from the model's inability to account for the attenuating effects of airborne particulate matter on PV module performance, which introduces uncertainty into irradiance reception and energy conversion processes. By comparison, the LSTM model demonstrates improved sensitivity to environmental perturbations and achieves a more accurate fitting of stochastic fluctuations under FECs.

Based on the above analysis, the challenges of PV power forecasting under FECs mainly fall into two categories: First, abrupt power fluctuations can lead to delayed or inaccurate predictions, placing higher

demands on the model's capacity to capture rapid variations. Second, stochastic fluctuations disrupt the stability of data distribution, and the inherent randomness of PV data further increases the difficulty of feature extraction and generalization modeling.

3. Methodology

This section presents the methodology of PKINN, beginning with an overview of how the proposed framework adaptively addresses different types of fluctuations. It then introduces the Quadratic Explicit Model (QEM) and the Fluctuation Allocation Mechanism (FAM), clarifying their roles in handling abrupt and stochastic fluctuations under FECs. To provide an overall view, Fig. 2 summarizes the end-to-end workflow of PKINN, illustrating the information flow from historical inputs through QEM and FAM to the corresponding prediction branches.

3.1. The structure of Photovoltaic Knowledge-Informed Neural Network (PKINN)

The structure of PKINN is illustrated in Fig. 3. The framework aims to address abrupt and stochastic power fluctuations in PV systems by analytical modeling with fluctuation quantification, thereby ensuring robust forecasting performance under diverse FECs. The PKINN consists of two components, QEM and FAM. QEM provides explicit characterization of abrupt fluctuations, while FAM performs adaptive modeling of stochastic fluctuations. In the PKINN framework, QEM employs an explicit quadratic formulation to characterize the mapping from environmental and electrical factors to PV power output, and constructs the current deviation function $D_{\text{cur}}(V)$ as its optimization objective. With explicit formulation and an iterative optimization strategy, QEM is able to capture abrupt fluctuations under FECs. The output of QEM serves as prior information for FAM, enabling FAM to concentrate on modeling stochastic fluctuations. FAM quantifies fluctuation intensity by introducing a fluctuation sensitivity coefficient λ and adapts sample routing according to different fluctuation levels [42]. By integrating fluctuation quantification with differentiated modeling, FAM provides a precise and adaptive representation of stochastic fluctuations.

To reconcile the modeling objectives of abrupt and stochastic fluctuations, the PKINN defines a loss function designed for complex fluctuation conditions. This loss function consists of Loss_{QEM} and Loss_{FAM} , where the former evaluates the model's prediction errors in capturing abrupt fluctuations through QEM, and the latter assesses its performance in representing stochastic fluctuations through FAM. The optimization objective is formulated as:

$$\text{Loss}_{\text{Total}} = e^{\lambda} \cdot \text{Loss}_{\text{QEM}} + \text{Loss}_{\text{FAM}} \quad (2)$$

where the term e^{λ} adaptively regulates the contribution of QEM relative to FAM under different fluctuation intensities. In PKINN, the fluctuation sensitivity coefficient λ modulates the contribution of the QEM loss through its exponential mapping e^{λ} , which adaptively adjusts the weight of the QEM constraint in the overall loss function. This design ensures that the loss function remains consistent with the characteristics of FECs. As a result, the coefficient λ is transformed into an always-positive weighting factor that continuously regulates the strength of Loss_{QEM} . Lower values of λ , including the negative regime, correspond to weaker QEM enforcement and increased emphasis on stochastic fluctuation learning, whereas higher values of λ lead to stronger QEM constraints, enhancing the analytical modeling of abrupt fluctuation conditions. In this way, the PKINN flexibly adjusts its forecasting strategies across diverse FECs, thereby improving interpretability and predictive performance. The pseudocode of the PKINN procedure is presented in Algorithm 1.

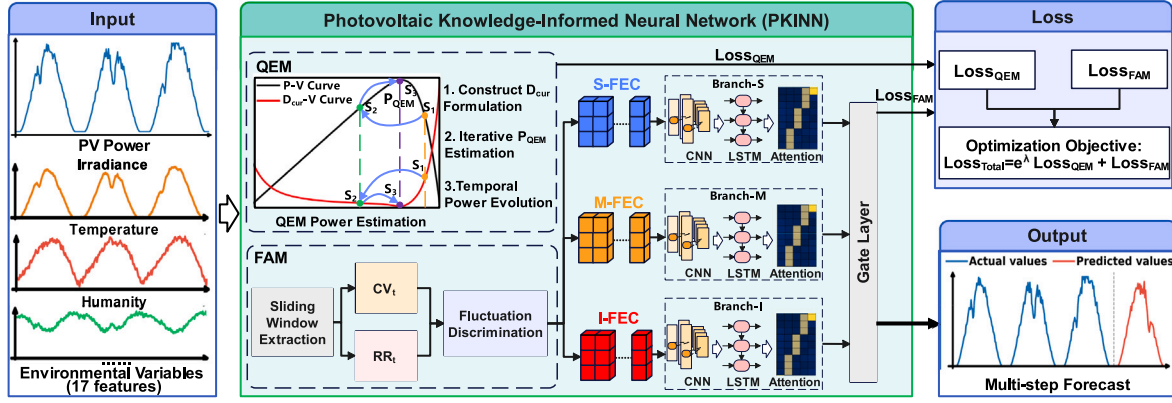


Fig. 2. End-to-end workflow of the PKINN framework.

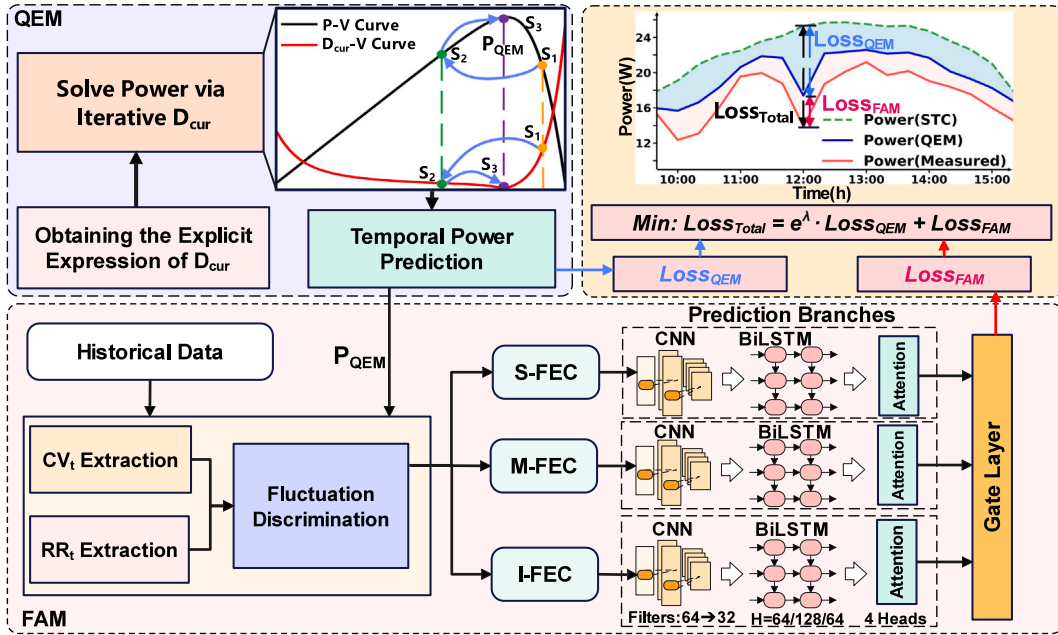


Fig. 3. Framework of the proposed PKINN.

3.2. Quadratic Explicit Model (QEM)

To address abrupt power fluctuations in PV systems, the PKINN framework proposes the QEM, which establishes explicit functional relationships among environmental variables, PV module electrical parameters, and power output. By reformulating the power equation into an algebraic quadratic form, QEM provides an interpretable mechanism for uncovering the intrinsic characteristics of power fluctuations. Specifically, QEM introduces a current deviation metric to quantify the discrepancy between the measured and reference currents, thereby enabling explicit characterization of the trajectory of convergence toward the target power point. This explicit formulation enhances the interpretability of the forecasting process. The loss function $Loss_{QEM}$ is given by:

$$Loss_{QEM} = \frac{1}{N} \sum_{i=1}^N D_{cur}(V_i). \quad (3)$$

where the current deviation function is defined as $D_{cur}(V_i) = |I_{emp}(V_i) - I_{ref}(V_i)|$, which represents the difference between the empirical current I_{emp} and the reference current I_{ref} at the sampling voltage V_i . As a quantitative metric evaluated over the voltage domain, D_{cur} measures the deviation between the measured current and the analytically

derived reference current. This formulation reformulates the power estimation process as a tractable single-variable optimization problem.

To obtain the reference current I_{ref} , QEM reformulates the single-diode model (SDM) into an explicit analytical form. This reformulation is carried out under steady-state operating assumptions, under which the PV module can be treated as a static nonlinear system over short time intervals, with spatially uniform cell temperature, homogeneous thermal distribution across the module, and negligible transient effects such as noise, parasitic capacitance. Under these conditions, the implicit nonlinear I-V relation can be algebraically reorganized into a quadratic expression with respect to the reference current, thereby transforming the original transcendental dependence into an explicitly analyzable quadratic form. To eliminate the exponential coupling between current and voltage and obtain a closed-form algebraic relation for the reference current, an additional analytical constraint is introduced by enforcing the condition $\partial P / \partial V = 0$. Combining this condition with the explicit I-V relation and the power derivative expression $\partial P / \partial V = V \partial I / \partial V + I$, the resulting constraint can be algebraically simplified into a quadratic equation with respect to the reference current, $k_0 I_{ref}^2 + k_1 I_{ref} + k_2 = 0$. Solving this quadratic equation yields a closed-form expression for the reference current:

$$I_{ref} = \frac{1}{2k_0} \left(-k_1 + \sqrt{k_1^2 - 4k_0k_2} \right), \quad (4)$$

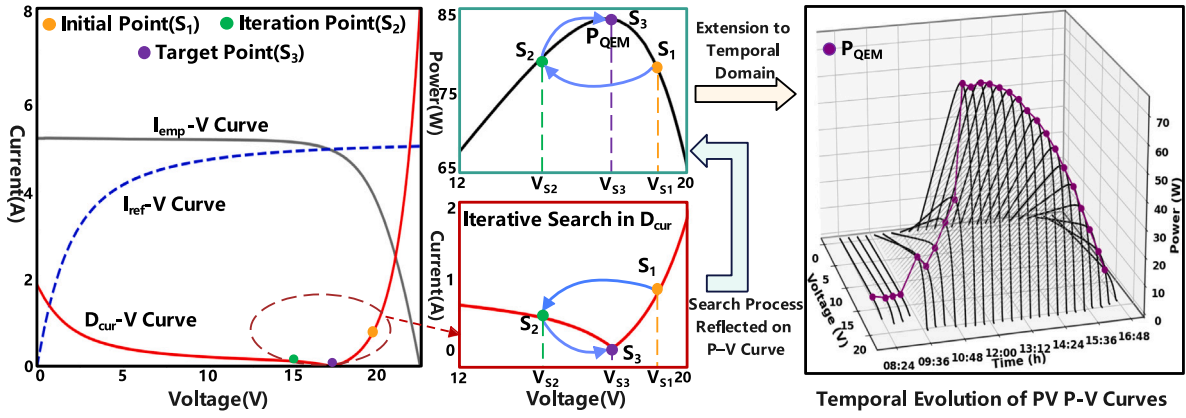


Fig. 4. QEM process for target power prediction.

Algorithm 1 Procedure of the PKINN Framework

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1: Input: Environmental variables  $\{E_t\}_{t=1}^T$ , measured power  $\{P_t\}_{t=1}^T$ ,
   QEM parameters  $\Theta$ , fluctuation thresholds  $(\alpha_t, \beta_t)$ , forecasting
   horizon  $H$ 
2: Output: Multi-step forecast  $\hat{P}_{t+1:t+H}$ 
3: Initialize branch network parameters and fluctuation coefficient  $\lambda$ ;
4: for each training epoch do
5:   QEM: Quadratic Explicit Model
6:   for  $t = 1$  to  $T$  do
7:     Extract environmental features from  $E_t$ ;
8:     Compute  $P_{QEM}(t)$  using the quadratic I-V model with param-
       eters  $\Theta$ ;
9:     Compute current deviation  $D_{cur}(t)$ ;
10:    Compute QEM loss as  $\text{Loss}_{QEM} = \frac{1}{N} \sum_{i=1}^N D_{cur}(V_i)$ ;
11:  end for
12:  FAM: Fluctuation Allocation Mechanism
13:  for  $t = L$  to  $T - H$  do
14:    Construct historical window  $\Psi_t = \{P_{t-L+1}, \dots, P_t\}$ ;
15:    Compute  $CV_t$  and  $RR_t$  from  $\Psi_t$ ;
16:    Determine fluctuation label  $F_t \in \{\text{S-FEC}, \text{M-FEC}, \text{I-FEC}\}$  using
       $(CV_t, RR_t, \alpha_t, \beta_t)$ ;
17:    Select prediction branch  $b \leftarrow F_t$ ;
18:    Form branch input by combining  $\Psi_t$ ,  $E_{t-L+1:t}$  and  $P_{QEM}(t)$ ;
19:    Obtain forecast  $\hat{P}_{t+1:t+H}^{(b)}$  through A-CNN-LSTM;
20:    Compute FAM loss as  $\text{Loss}_{FAM} = \frac{1}{N} \sum_{n=1}^N (\hat{y}_n - y_n)^2$ ;
21:  end for
22:  Compute total loss:  $\text{Loss}_{Total} = e^\lambda \text{Loss}_{QEM} + \text{Loss}_{FAM}$ 
23:  Update  $\lambda$  and all branch parameters via back-propagation;
24: end for
25: Return:  $\hat{P}_{t+1:t+H}$ 

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where the quadratic coefficient is $k_0 = -R_s(R_s + R_{sh})$, the linear coefficient is $k_1 = (I_{sc} + I_0)R_s R_{sh} + (AV_t + V)(R_s + R_{sh}) - R_s V$, and the constant term is $k_2 = V^2 - [(I_0 + I_{sc})R_{sh} + AV_t]V$. In these expressions, R_s and R_{sh} denote the series and shunt resistances, V_t is the thermal voltage, A is the diode ideality factor, I_0 is the reverse saturation current, and I_{sc} is the short-circuit current. These coefficients depend on the electrical parameters (R_s , R_{sh} , V_t), current-related terms (I_0 , I_{sc}), and the operating voltage V , thereby transforming the nonlinear I-V relation into an analytically tractable quadratic form [43]. The resulting I_{ref} represents the analytically derived reference current corresponding to the ideal operating condition and serves as an analytically defined target that guides power estimation within QEM.

The forecasting process of PV power in QEM is illustrated in Fig. 4. QEM employs PV module parameters to determine the initial power

point P_{STC} under Standard Test Conditions (STC), which serves as the starting point of the search process. The parameters of the PV module are then extracted from historical I-V curves through a two-step linear least squares method. With these parameters, the model iteratively applies the current deviation loss function, ultimately identifying the analytically predicted power point P_{QEM} together with its corresponding current and voltage values. As environmental conditions vary, PV module parameters dynamically change, leading to fluctuations in PV power output [44]. As shown in Fig. 4, QEM characterizes power variations under different operating conditions by leveraging the explicit analytical relationships between PV modules and their environments. The power estimated by QEM, denoted as P_{QEM} , is subsequently introduced into the downstream modules of PKINN, providing explicit analytical guidance for learning stochastic fluctuations while preserving sensitivity to abrupt power variations. Moreover, the framework ensures analytical consistency of the prediction results, which enhances the efficiency and stability of the forecasting strategy. The curve of the current deviation function exhibits a unique global optimum within the defined solution interval, ensuring stability throughout the prediction process. During training, PV power computation is integrated with model parameter optimization, allowing the analytical formulation to guide the optimization trajectory from the outset. This explicit modeling approach maintains consistency with fluctuation analysis principles, reducing reliance on large-scale training data and improving computational efficiency. The complete derivation and dimensional verification of the QEM are provided in Appendix.

3.3. Fluctuation Allocation Mechanism (FAM)

In PV power forecasting, environmental conditions are often affected by random disturbances, which lead to stochastic fluctuations in power output. These fluctuations are difficult to capture and pose considerable challenges to forecasting accuracy and generalization. To address this issue, the PKINN introduces FAM, which explicitly quantifies fluctuation intensity and adaptively allocates input data to specialized prediction branches. A fluctuation sensitivity coefficient λ is further employed to dynamically regulate the thresholding process, allowing flexible distinction among different FECs.

The fluctuation condition is evaluated in a time-dependent manner using a sliding historical window, where the corresponding set of historical samples up to time t is denoted as $\Psi_t = \{y_{t-L+1}, y_{t-L+2}, \dots, y_t\}$, with L representing the window length and Ψ_t containing all past observations within this window. Within Ψ_t , the fluctuation intensity is quantified by two statistical indicators: the coefficient of variation (CV) and the ramp rate (RR) [45]. The CV is defined as $CV_t = \frac{\sigma(\Psi_t)}{\mu(\Psi_t) + \epsilon}$, where σ and μ denote the standard deviation and mean of the set Ψ_t , respectively, and ϵ is a small constant introduced to avoid division by zero [46]. This indicator characterizes the relative amplitude of

variations by normalizing the standard deviation by the mean, which reduces the influence of scale differences among variables. The RR is defined as:

$$RR_t = \frac{1}{L-1} \sum_{y_i \in \mathcal{Y}_t} \frac{|y_i - y_{i-1}|}{P_{rated}}, \quad (5)$$

which represents the local average rate of change and quantifies short-term ramping dynamics from the differences between adjacent time steps.

Based on these two indicators, the fluctuation condition label F_t is adaptively determined at each time step to both characterize the current fluctuation regime and guide the allocation of samples to corresponding prediction branches. Specifically, F_t identifies whether the system is under a stable (S-FEC), moderate (M-FEC), or intense (I-FEC) fluctuation state according to the following rule:

$$F_t = \begin{cases} \text{S-FEC,} & \text{if } CV_t \leq \overline{CV} e^{\alpha_1 \lambda} \text{ or } RR_t \leq \overline{RR} e^{\beta_1 \lambda}, \\ \text{I-FEC,} & \text{if } CV_t \geq \overline{CV} e^{\alpha_2 \lambda} \text{ or } RR_t \geq \overline{RR} e^{\beta_2 \lambda}, \\ \text{M-FEC,} & \text{otherwise.} \end{cases} \quad (6)$$

where $\overline{CV}$ and $\overline{RR}$ denote the mean values of CV and RR computed from the training dataset, serving as normalized references for threshold determination. The parameters α_i and β_i control the sensitivity of the thresholds, typically within the range (0.05–0.5) [47]. The coefficient λ is embedded in the exponential functions to stretch or contract the thresholds and thereby adaptively allocate samples to the corresponding prediction branches.

After allocation, the fluctuation condition label F_t explicitly determines which prediction branch is activated at each time step. The prediction branch corresponding to the identified fluctuation regime is responsible for forward prediction and parameter updating. Formally, the branch selection can be represented by a routing vector $\mathbf{r}_t \in \{0, 1\}^3$, defined as:

$$\mathbf{r}_t = \begin{cases} [1, 0, 0], & F_t = \text{S-FEC,} \\ [0, 1, 0], & F_t = \text{M-FEC,} \\ [0, 0, 1], & F_t = \text{I-FEC.} \end{cases} \quad (7)$$

which establishes the correspondence between the identified fluctuation condition and the activated prediction branch. Under S-FEC, the prediction branch focuses on stable trends and irradiance information while capturing slight fluctuation patterns. Under M-FEC, the prediction branch emphasizes moderate variations as well as stochastic fluctuation features. Under I-FEC, the branch concentrates on rapid ramping behaviors and higher-order fluctuation patterns, leveraging the abrupt features indicated by QEM to capture sudden changes. Each prediction branch is implemented with an Attention-CNN-LSTM (A-CNN-LSTM) network. Within the A-CNN-LSTM, the CNN module is first employed to extract short-term disturbance features and reduce feature dimensionality by capturing local temporal patterns. Subsequently, the Bi-LSTM module effectively models long-term and bidirectional temporal dependencies, enabling the network to learn dynamic variations in the power sequence. Furthermore, an attention mechanism is integrated to adaptively weight historical time steps, highlighting those most relevant to fluctuation dynamics [48]. To enforce the fluctuation-aware branch selection within the network architecture, a gating layer is introduced as a structural selector following the parallel prediction branches. The gating layer functions as a structural selector that realizes the branch activation determined by the fluctuation condition label F_t . Specifically, it generates a branch-wise activation mask corresponding to F_t , allowing the output of the selected prediction branch to be propagated to the final output. Through this mechanism, discrete branch activation is consistently enforced under continuously varying input conditions, ensuring stable and coherent branch selection across different FECs.

FAM quantifies different fluctuation intensities through λ and adaptively allocates data to prediction branches corresponding to different fluctuation intensities, where dedicated networks learn the features of slight, moderate, and intense fluctuations. The loss function of FAM is defined as:

$$\text{Loss}_{\text{FAM}} = \frac{1}{N} \sum_{n=1}^N (\hat{y}_n - y_n)^2, \quad (8)$$

where y_n denotes the observed power, $\hat{y}_n$ prediction result, and N is the number of samples. This loss quantifies the deviation between predicted and true values under the fluctuation-aware allocation mechanism. During training, the loss is back-propagated through the prediction branch selected by F_t , enabling each branch to be optimized with samples associated with its designated fluctuation regime. By identifying fluctuation intervals and facilitating targeted optimization, FAM enhances both the stability and interpretability of PV power prediction under diverse FECs.

4. Experimental results and discussion

This section evaluates the prediction performance of the proposed PKINN under FECs. Section 4.1 introduces the experimental data and setup. Section 4.2 presents ablation experiments to examine the respective roles and contributions of QEM, FAM, and the fluctuation sensitivity coefficient λ . Section 4.3 compares the PKINN with several State-Of-The-Art (SOTA) forecasting models to assess its accuracy. Section 4.4 further investigates the model's performance under different fluctuation intensities, analyzing how prediction errors and the value of λ vary across FECs.

4.1. Experimental data and setup

The PV data used in this study were sourced from the National Renewable Energy Laboratory (NREL) database [49]. Four PV module cases were selected to evaluate the generalization capability of the forecasting model:

- **xSi12922 (Case 1):** A monocrystalline silicon (x-Si) module with a rated power of 83.04 W, an approximate conversion efficiency of 20%, and 36 series-connected cells. The data were collected in Cocoa, Florida (28.39°N, 80.46°W), and the testing data used measurements from September to December.
- **xSi12922 (Case 2):** The module parameters are identical to Case 1. The data were collected in Eugene, Oregon (44.05°N, 123.07°W), and the testing data used measurements from January to March.
- **mSi0166:** A multicrystalline silicon (m-Si) module with a rated power of 46.08 W, an approximate conversion efficiency of 16%, and 36 series-connected cells. The data were collected in Cocoa, Florida (28.39°N, 80.46°W), and the testing data used measurements from September to December.
- **HIT05667:** A heterojunction (HIT) module with a rated power of 245.10 W, an approximate conversion efficiency of 17%, and 60 series-connected cells. The data were collected in Cocoa, Florida (28.39°N, 80.46°W), and the testing data used measurements from September to December.

These four cases collectively cover different silicon-based PV technologies, power capacities, locations, and seasonal conditions, enabling a comprehensive evaluation under diverse operating scenarios.

The dataset covers a complete one-year observation period and contains approximately 17,000 samples per module. Each dataset includes 18 feature variables, encompassing meteorological factors such as irradiance and temperature, as well as relevant PV module parameters. The data were recorded at a 5-minute sampling interval in the NREL dataset, which provides adequate temporal resolution to

Table 1

Architectural details of the A-CNN-LSTM prediction branch in the PKINN.

Layer	Layer type	Configuration	Output size
Input	Input layer	–	72×18
CNN	Convolution layer	Kernel = 3, Filters = 64	72×64
CNN	Max-pooling layer	Pool size = 2	36×64
CNN	Convolution layer	Kernel = 3, Filters = 32	36×32
Bi-LSTM	Bidirectional LSTM layer	64 units	36×128
Bi-LSTM	Bidirectional LSTM layer	128 units	36×256
Bi-LSTM	Bidirectional LSTM layer	64 units	36×128
Attention	Multi-head attention layer	4 heads	36×128
Output	Dense output layer	Linear	128

Note: The output size is reported in the form of (time steps $\times$ feature dimension). The total number of trainable parameters is 3.2×10^5 , and the computational complexity is on the order of $\mathcal{O}(10^6)$ FLOPs per forward pass.

capture the fluctuations present in PV power output [50]. The PV and environmental data were cleaned and filtered by retaining only daytime records between 08:00 and 17:00 and removing nighttime samples with zero power output. Abnormal observations caused by sensor drift or meteorological measurement errors were identified and removed, while missing values due to sensor faults were imputed using a time-based polynomial interpolation method to ensure continuity of the data sequence. After preprocessing, structured samples were constructed for the one-hour forecasting task, which is a commonly adopted horizon in short-term PV prediction, and the data were divided into training, validation, and testing sets using a ratio of 8:1:1 [51].

All models were implemented in Python using TensorFlow 2.9 and trained on an NVIDIA RTX 3090 GPU. The Adam optimizer was employed with an initial learning rate of 0.001, and early stopping was applied with a patience of 20 epochs. For the proposed PKINN, the FAM adopted scaling parameters of $\alpha_1 = 0.15$, $\alpha_2 = 0.10$, $\beta_1 = 0.25$, and $\beta_2 = 0.05$, which were empirically determined on the validation set to balance sensitivity across different levels of FECs. The detailed architecture of the A-CNN-LSTM prediction branch used in PKINN is summarized in Table 1. The table provides a specification of the network configuration, including convolutional layers, bidirectional LSTM layers, and the multi-head attention module, together with their corresponding output dimensions. All models used for experimental comparison were trained for up to 200 epochs with a batch size of 64.

Model performance was evaluated using three commonly employed point-forecasting metrics, namely Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). These metrics are expressed as $\text{RMSE} = \sqrt{\frac{1}{N} \sum_{t=1}^N (\hat{y}_t - y_t)^2}$, $\text{MAE} = \frac{1}{N} \sum_{t=1}^N |\hat{y}_t - y_t|$, and $R^2 = 1 - \frac{\sum_{t=1}^N (\hat{y}_t - \bar{y})^2}{\sum_{t=1}^N (y_t - \bar{y})^2}$. In addition to point-forecasting accuracy, two uncertainty-related metrics were adopted: the Prediction Interval Normalized Average Width (PINAW) and the Continuous Ranked Probability Score (CRPS) [52], defined as:

$$\text{PINAW} = \frac{1}{N(Y_{\max} - Y_{\min})} \sum_{t=1}^N (U_t - L_t). \quad (9)$$

$$\text{CRPS} = \frac{1}{N} \sum_{t=1}^N \int_{-\infty}^{+\infty} [P_t(y) - H(y - y_t)]^2 dy. \quad (10)$$

where $\hat{y}_t$ and y_t denote the predicted and measured PV power, $\bar{y}$ is the sample mean of the observations, U_t and L_t denote the upper and lower prediction-interval bounds, $P_t(y)$ is the cumulative distribution function of the predictive distribution, and H is the Heaviside step function.

4.2. Verification of QEM mechanism and ablation analysis

This section verifies the effectiveness of the proposed PKINN framework and analyzes the contribution of each key module through ablation experiments. The evaluation begins by demonstrating the capability of QEM to capture the target power point under both abrupt

and stochastic fluctuations. Subsequently, the impact of progressively removing the components of the framework, including QEM, FAM, and λ , is quantitatively assessed across multiple PV module datasets and evaluation metrics. Fig. 5 presents the P_{QEM} search trajectories under two representative fluctuation conditions, including an abrupt fluctuation case and a stochastic fluctuation case. Each subfigure displays the measured power sequence, the power–voltage curve, the voltage-dependent reference current I_{ref} , and the voltage-dependent current deviation D_{cur} , along with the corresponding QEM search trajectory. During the search process, QEM computes the reference current I_{ref} from the voltage-dependent characteristics and evaluates the current deviation D_{cur} . The current deviation serves as the criterion for identifying the target power point and guiding the search direction. Starting from the STC as the initial operating point, the QEM search iteratively updates the operating voltage and moves along an analytically determined search path toward the target power point. In both fluctuation scenarios, the search trajectory converges to the voltage at which the current deviation becomes zero, corresponding to the condition that the slope of the power–voltage curve vanishes. These experimental results demonstrate that the reference current I_{ref} and the current deviation D_{cur} together provide an interpretable guidance mechanism for determining P_{QEM} under FECs.

To further assess the contribution of each module, three ablation variants were constructed: PKINN-w/o-QEM (removing QEM), PKINN-w/o-FAM (removing FAM), and PKINN-w/o- λ (removing λ). Two baseline models, EAM and LSTM, were also included for comparison. Table 2 summarizes the performance across four PV module datasets using five evaluation metrics, with the best results highlighted in bold. As shown in Table 2, across all four PV module cases, PKINN consistently achieves the best performance in terms of both point-forecasting accuracy and uncertainty metrics. On the xSi12922 dataset, PKINN reduces RMSE by more than 4.7 W relative to EAM and LSTM under both cases. On the mSi0166 dataset, the RMSE reduction consistently exceeds 2.3 W, while on the large-capacity HIT05667 module, the performance gains become substantially larger, indicating an advantage of PKINN under strong fluctuation conditions. These results suggest that baseline models relying solely on physical modeling or data-driven learning struggle to cope with the coexistence of stochastic and abrupt fluctuations under FECs. Among the three ablated variants, PKINN-w/o-QEM exhibits the most notable degradation, with RMSE increases of 30%–45% across different datasets. PKINN-w/o-FAM also shows notable degradation, with RMSE increases on the order of 20%–40%. Although PKINN-w/o- λ remains relatively stable, its RMSE is still 4.1%–22.2% higher than that of the complete PKINN across all module cases, accompanied by less favorable PINAW and CRPS values. These results demonstrate that QEM, FAM, and the adaptive λ contribute to the robustness and high accuracy of PKINN under diverse fluctuation conditions.

Fig. 6 presents a visual comparison of the prediction performance between PKINN and its ablated variants on the xSi12922 dataset, including the prediction curves under three types of FECs, as well as bar charts comparing the RMSE, MAE, and R^2 metrics under each fluctuation condition. PKINN-w/o-QEM shows significant deviations during abrupt fluctuations, suggesting that the absence of QEM limits its ability to capture sudden power changes. By contrast, PKINN-w/o-FAM also shows error accumulation under M-FEC and I-FEC, indicating that the lack of a fluctuation allocation mechanism weakens adaptability across different fluctuation intensities. PKINN-w/o- λ delivers relatively more stable performance, but the absence of adaptive regulation reduces its generalization across FECs, causing its RMSE and MAE to remain higher than those of the complete PKINN. Fig. 6 demonstrates that removing QEM, FAM, or adaptive λ leads to progressively larger prediction errors compared with the complete PKINN, confirming the contributions of these components to robust performance.

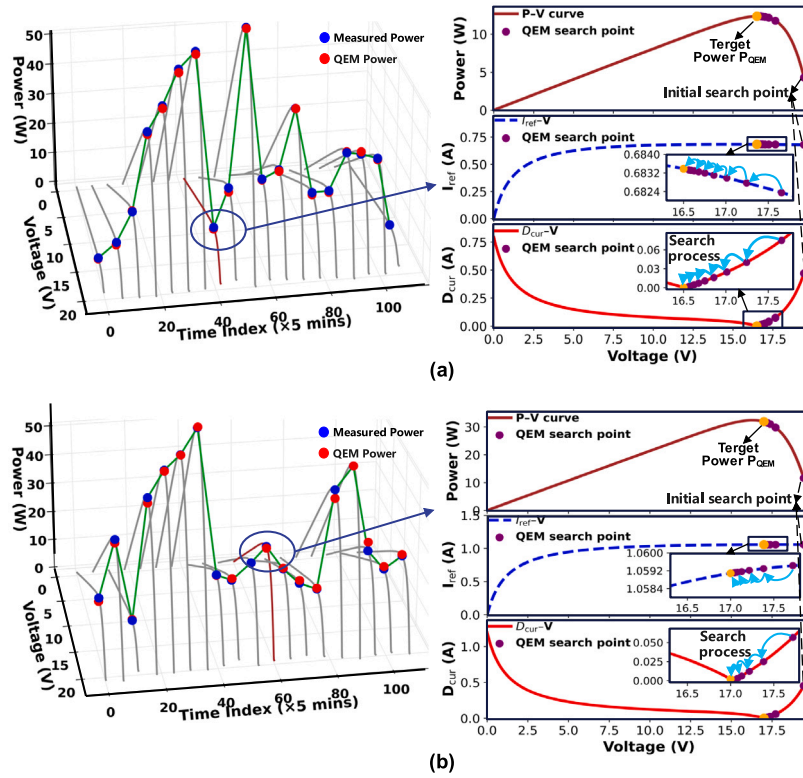


Fig. 5. Power search process using QEM for (a) an abrupt fluctuation case and (b) a stochastic fluctuation case.

Table 2

Performance comparison across PV module cases using baseline and ablation models.

PV Module type	Metric	EAM [11]	LSTM [24]	PKINN-w/o-QEM	PKINN-w/o-FAM	PKINN-w/o- λ	PKINN
xSi12922 (83 W) Case 1	RMSE (W)	12.57	11.94	9.53	8.72	7.64	7.24
	MAE (W)	9.26	8.31	7.09	6.57	5.79	5.45
	R ² (%)	74.57	77.29	82.72	85.93	87.41	91.42
	PINAW	0.63	0.59	0.47	0.39	0.50	0.29
	CRPS	9.99	10.85	8.71	7.04	6.69	4.46
xSi12922 (83 W) Case 2	RMSE (W)	11.19	10.26	7.53	7.06	6.05	5.53
	MAE (W)	8.99	6.07	5.29	5.03	4.60	3.89
	R ² (%)	76.28	77.64	83.81	86.59	89.30	92.83
	PINAW	0.53	0.53	0.39	0.37	0.28	0.21
	CRPS	7.93	7.43	5.70	4.98	3.33	2.28
mSi0166 (46 W)	RMSE (W)	8.26	7.38	6.74	6.12	5.34	5.13
	MAE (W)	5.81	5.38	4.11	4.04	3.45	3.25
	R ² (%)	74.76	78.50	83.33	86.06	90.54	93.82
	PINAW	0.59	0.60	0.44	0.39	0.37	0.30
	CRPS	6.90	6.19	4.51	4.02	3.82	2.45
HIT05667 (245 W)	RMSE (W)	34.37	31.19	22.64	22.56	19.28	15.78
	MAE (W)	26.09	24.91	17.38	17.01	14.39	13.39
	R ² (%)	70.80	71.50	80.40	81.40	84.50	88.40
	PINAW	0.34	0.69	0.45	0.43	0.54	0.29
	CRPS	26.71	24.78	20.14	18.26	16.12	11.54

4.3. Comparative evaluation with SOTA models

To evaluate the predictive performance of the proposed model, Table 3 compares PKINN with five representative SOTA time-series forecasting models, including A-CNN-LSTM [53], TimesNet [54], SegRNN [55], Informer [56], and Transformer [57]. For the two testing cases in the xSi12922 dataset (Case 1 and Case 2), Case 2 exhibits weaker irradiance and smaller power variations, resulting in lower errors than Case 1. PKINN maintains superiority in both cases, achieving RMSE reductions of 17.5%–22.8% in Case 1 and 12.4%–31.2% in Case 2, together with the lowest MAE and highest R^2 . These results demonstrate that PKINN consistently outperforms the SOTA models across different locations and seasonal conditions. On the mSi0166

dataset, which has a lower rated power and a smaller proportion of high-fluctuation samples, all models show reduced errors. Nevertheless, PKINN remains the best-performing model, yielding RMSE improvements of 9.8%–18.1% compared with other SOTA models. For the HIT05667 dataset, PKINN exhibits an even more substantial advantage, achieving RMSE reductions of 15.6%–36.5% over the five competing models. This indicates that PKINN can effectively adapt to the fluctuation characteristics of large-capacity PV modules. Regarding uncertainty quantification, PKINN achieves the lowest PINAW and CRPS, indicating that its prediction intervals are not only narrower and better calibrated but also provide a more faithful representation of the underlying predictive distribution. Among the comparison models, SegRNN provides relatively strong baseline performance due to its

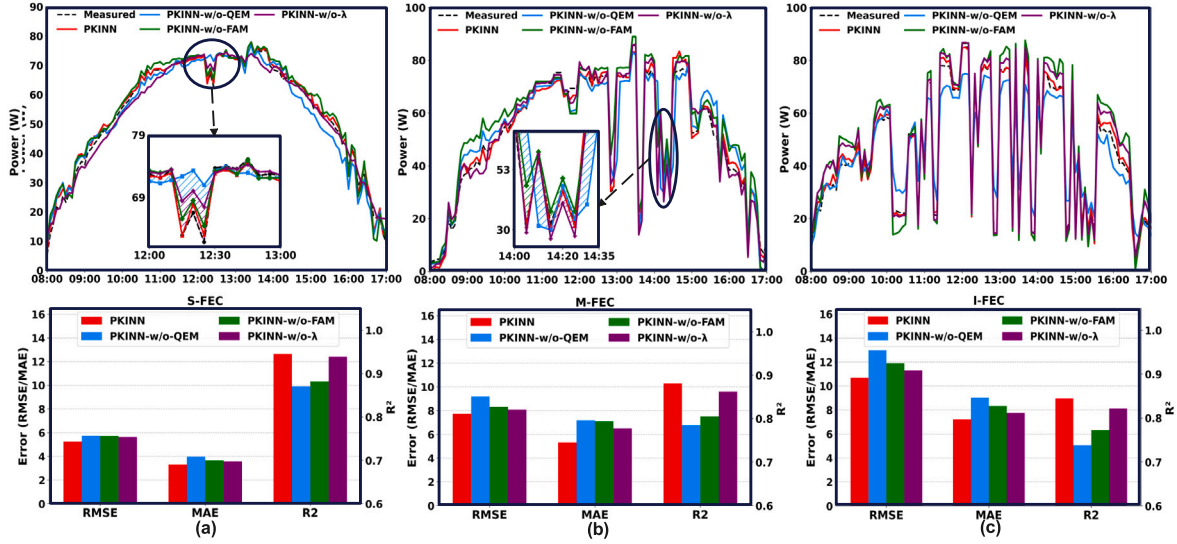


Fig. 6. Performance comparison of PKINN and Ablation models under different FECs: (a) S-FEC, (b) M-FEC, (c) I-FEC.

Table 3

Performance comparison across PV module cases using SOTA forecasting models.

PV Module type	Metric	A-CNN-LSTM [53]	TimesNet [54]	SegRNN [55]	Informer [56]	Transformer [57]	PKINN
xSi12922 (83 W) Case 1	RMSE (W)	8.78	8.22	7.93	8.38	9.37	7.24
	MAE (W)	6.71	6.32	5.99	6.94	7.18	5.45
	R ² (%)	83.51	84.88	85.65	83.95	81.89	91.42
	PINAW	0.55	0.51	0.43	0.53	0.49	0.29
	CRPS	9.12	8.07	6.80	8.71	8.99	4.46
xSi12922 (83 W) Case 2	RMSE (W)	7.39	6.57	6.17	6.86	8.04	5.53
	MAE (W)	4.95	4.49	4.34	4.94	5.35	3.89
	R ² (%)	81.02	84.93	85.67	82.90	79.71	92.83
	PINAW	0.39	0.34	0.29	0.35	0.36	0.21
	CRPS	5.84	5.06	3.99	5.60	5.61	2.28
mSi0166 (46 W)	RMSE (W)	5.96	5.73	5.69	5.90	6.26	5.13
	MAE (W)	3.90	3.53	3.59	3.86	4.17	3.25
	R ² (%)	86.23	87.35	89.15	85.09	84.09	93.82
	PINAW	0.53	0.42	0.40	0.48	0.47	0.30
	CRPS	5.23	4.15	3.84	4.52	4.52	2.45
HIT05667 (245 W)	RMSE (W)	22.68	19.29	18.71	22.37	24.85	15.78
	MAE (W)	17.65	14.34	14.18	18.87	20.18	13.39
	R ² (%)	78.80	80.90	82.10	77.90	74.30	88.40
	PINAW	0.64	0.43	0.39	0.53	0.52	0.29
	CRPS	22.79	18.21	15.95	23.44	24.23	11.54

multi-period spatiotemporal coupling structure, whereas Informer is hindered by its sparse attention design, leading to weaker performance under rapidly fluctuating conditions. In contrast, Transformer performs the worst, as its limited capability in modeling temporal dependencies reduces its ability to capture short-term PV fluctuations, resulting in substantial errors under FECs.

Fig. 7 presents the power prediction results across four PV modules, including xSi12922 (Case 1 and Case 2), mSi0166, and HIT05667. It provides an intuitive comparison between PKINN and SOTA models under complex FECs, highlighting their responses to stochastic and abrupt fluctuations across different locations, seasons, and power scales. As shown in Fig. 7(a) and (b), which correspond to the xSi12922 module under different locations and seasons, abrupt fluctuations pose significant challenges for most SOTA models. A-CNN-LSTM and Informer exhibit noticeable prediction delays and amplitude deviations. SegRNN is able to follow the trend but shows instability under sudden power fluctuations, while TimesNet demonstrates stronger responsiveness owing to its time-frequency fusion structure. In contrast, PKINN consistently produces the most accurate prediction results, capturing both the temporal dynamics and fluctuation amplitudes. Moreover, Fig. 7(c) and (d) further illustrate prediction behaviors across PV modules

with different capacities and electrical characteristics. Under stochastic fluctuations, A-CNN-LSTM exhibits the most pronounced deviations, struggling to cope with random variations. By comparison, PKINN and SegRNN show greater robustness in tracking stochastic fluctuations across different PV modules, with PKINN maintaining closer alignment with the measured power.

Fig. 8 presents the scatter plots of predicted versus measured values for each model, along with the 90% prediction intervals obtained from linear regression. PKINN exhibits the narrowest interval and a regression line closely aligned with the ideal 1:1 reference, indicating strong capability in characterizing both abrupt and stochastic fluctuations. Most scatter points of SegRNN and TimesNet fall within the prediction intervals, suggesting relatively stable performance under I-FECs. By contrast, Transformer shows a more dispersed distribution, with numerous points lying outside the interval, reflecting lower accuracy, particularly under conditions of pronounced power fluctuations. Moreover, A-CNN-LSTM and Informer also display a portion of out-of-band points, which indicates their insufficient responsiveness to abrupt fluctuations.

To further analyze the error characteristics, Fig. 9 provides a comparison of absolute error distributions and residual density curves

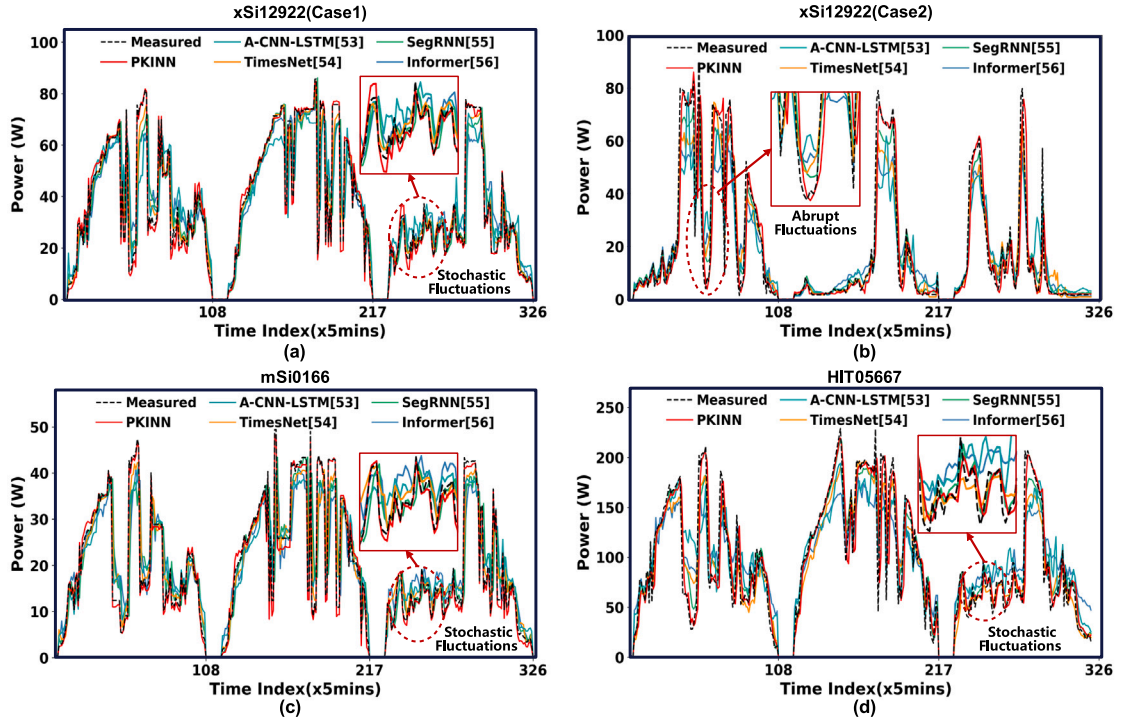


Fig. 7. Comparison of PV power prediction results for PKINN and representative SOTA models across different PV modules: (a) xSi12922 (Case 1), (b) xSi12922 (Case 2), (c) mSi0166, and (d) HIT05667.

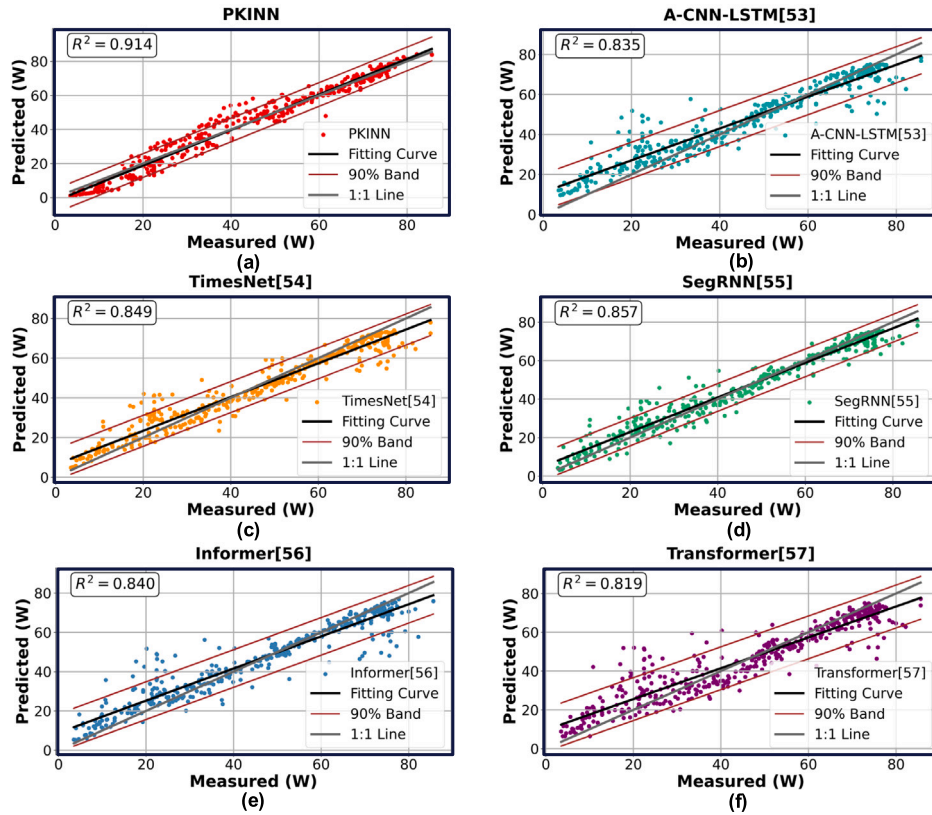


Fig. 8. Scatter plots of model predictions for (a) PKINN, (b) A-CNN-LSTM [53], (c) TimesNet [54], (d) SegRNN [55], (e) Informer [56], and (f) Transformer [57]. Each subplot includes a fitted regression line and a 90% prediction band.

across models. PKINN achieves the most compact absolute error distribution, with significantly fewer large-error outliers than the SOTA models. The residual KDE curves show that PKINN produces a sharper

distribution centered near zero, indicating higher calibration quality and reduced systematic bias. In contrast, Transformer and Informer display wider and heavier-tailed residual distributions, revealing their

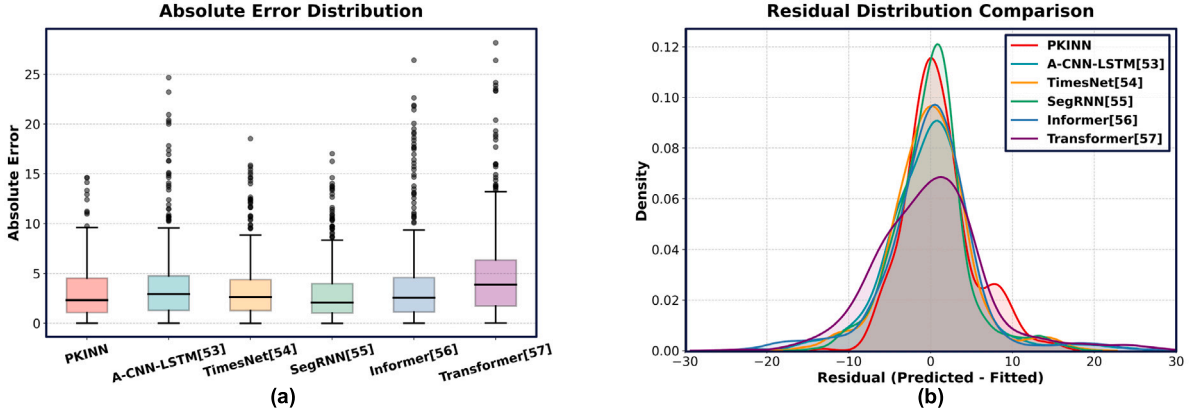


Fig. 9. Uncertainty analysis of prediction results: (a) absolute error distribution of different models, (b) residual distribution comparison (KDE curves).

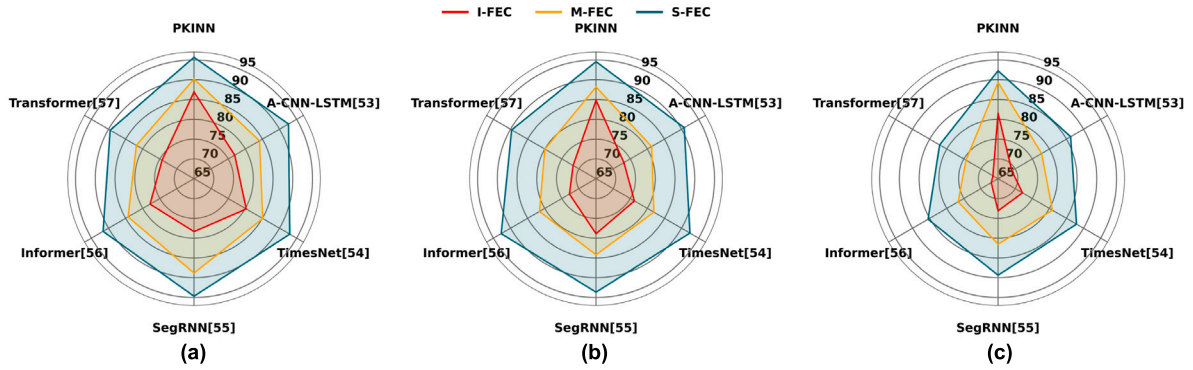


Fig. 10. Radar charts of R^2 performance for different models under varying FECs on three PV modules: (a) xSi12922, (b) mSi0166, and (c) HIT05667.

sensitivity to high-frequency fluctuations. SegRNN exhibits relatively moderate error behavior, but still shows a noticeably broader spread compared with PKINN. These visual results confirm that PKINN delivers more reliable and stable residual behavior under fluctuating conditions.

4.4. Prediction performance under different FECs

To comprehensively evaluate the predictive performance of different models under FECs, Table 4 summarizes the comparative results of PKINN and several SOTA models under S-FEC, M-FEC, and I-FEC, together with the corresponding fluctuation sensitivity coefficient λ for each case. Fig. 10 presents radar charts of R^2 values across the three FECs, providing a more intuitive comparison of model performance. Overall, all models achieve their highest prediction accuracy under S-FEC. However, as fluctuation intensity increases, model performance begins to diverge significantly. Specifically, SegRNN outperforms other SOTA models under both S-FEC and M-FEC, demonstrating its advantage in mild fluctuation conditions. Moreover, as shown in Fig. 10, SegRNN and TimesNet exhibit relatively minor degradation in R^2 with increasing fluctuation intensity. By contrast, A-CNN-LSTM experiences a marked performance decline under I-FEC, with its RMSE increasing by 4.72 on the xSi12922 (Case 1) dataset and 2.60 on the mSi0166 dataset compared to M-FEC. The Transformer model performs the worst across FECs, consistently yielding higher prediction errors than the other models. For the HIT05667 module, which has a higher rated capacity and exhibits stronger power variability, the performance divergence becomes even more pronounced. Under M-FEC and I-FEC, both Informer and Transformer show significant error inflation. These

results illustrate that large-capacity PV modules pose greater challenges for forecasting models.

The proposed PKINN achieves almost optimal performance under all FECs across both datasets. Benefiting from its interpretable architecture, PKINN enhances trend modeling through QEM and employs FAM to allocate and process fluctuations of different intensities, strengthening its ability to quantify and identify fluctuation features. As shown in Table 4, the learned fluctuation-sensitivity term e^{λ} increases steadily from S-FEC to M-FEC and I-FEC, reflecting the rising fluctuation intensity. Under I-FEC, a larger e^{λ} directs more training data to the branches specialized for strong fluctuations, enabling the model to better capture abrupt power fluctuations. Under M-FEC, by contrast, FAM places greater emphasis on suppressing the interference caused by random fluctuations. Across all datasets, PKINN achieves the highest R^2 , the lowest RMSE and MAE, and the narrowest prediction intervals (PINAW), together with the lowest CRPS values. Under I-FEC, PKINN improves R^2 by approximately 9.25% and 8.76% over the best-performing SOTA models on xSi12922 (Case 1) and mSi0166, respectively, and by 8.38% on the large-capacity HIT05667 module. These results, along with the narrow prediction intervals illustrated in Fig. 8, provide strong evidence of PKINN's robustness and adaptability under complex FECs.

5. Conclusion

This paper presents an interpretable PKINN framework to enhance PV power forecasting under FECs. The PKINN model specifically addresses the challenges of abrupt and stochastic fluctuations. The proposed QEM establishes explicit expressions of power variations and

Table 4
Comparison of model performance under different levels of FECs.

Dataset	Condition	$e^2(\times 10^{-3})$	Metric	A-CNN-LSTM [53]	TimesNet [54]	SegRNN [55]	Informer [56]	Transformer [57]	PKINN
xSi12922 (Case 1) (83 W)	S-FEC	6.3	RMSE (W)	5.41	5.33	5.12	5.34	5.82	5.23
			MAE (W)	3.79	3.59	3.57	3.64	4.10	3.31
			R ² (%)	90.79	92.46	93.66	92.74	89.67	94.56
			PINAW	0.46	0.32	0.21	0.36	0.37	0.19
			CRPS	6.87	5.05	4.22	6.89	5.70	3.37
	M-FEC	12.1	RMSE (W)	8.96	8.87	8.21	8.89	9.22	7.75
			MAE (W)	7.23	6.99	6.23	7.06	7.55	5.53
			R ² (%)	81.26	81.91	84.21	81.54	77.96	88.17
			PINAW	0.56	0.56	0.49	0.59	0.55	0.28
			CRPS	9.05	7.43	6.85	9.70	9.13	4.47
	I-FEC	21.5	RMSE (W)	13.68	12.26	11.90	13.87	14.58	10.28
			MAE (W)	9.24	8.91	8.10	9.52	10.26	7.02
			R ² (%)	73.15	76.23	78.96	72.77	71.71	86.20
			PINAW	0.68	0.73	0.62	0.66	0.75	0.36
			CRPS	12.18	10.28	9.33	11.69	11.33	5.77
mSi0166 (46 W)	S-FEC	8.2	RMSE (W)	2.89	2.72	2.61	3.01	3.44	2.51
			MAE (W)	2.61	2.30	2.33	2.73	2.99	1.90
			R ² (%)	92.64	93.00	94.69	91.60	89.52	95.68
			PINAW	0.31	0.26	0.25	0.29	0.26	0.18
			CRPS	3.05	2.28	2.14	2.95	2.21	1.72
	M-FEC	16.3	RMSE (W)	6.01	5.85	5.68	5.94	6.44	5.41
			MAE (W)	4.03	3.85	3.88	3.94	4.42	3.81
			R ² (%)	84.16	85.09	87.83	84.28	79.86	90.18
			PINAW	0.57	0.59	0.48	0.56	0.46	0.28
			CRPS	5.13	5.41	5.36	5.33	5.52	2.45
	I-FEC	23.0	RMSE (W)	8.61	7.71	7.88	8.20	9.11	6.35
			MAE (W)	6.29	5.57	5.79	5.84	6.89	4.98
			R ² (%)	76.91	80.28	78.36	77.86	74.34	87.32
			PINAW	0.65	0.65	0.61	0.66	0.57	0.31
			CRPS	6.90	7.09	6.31	6.90	7.46	3.14
HIT05667 (245 W)	S-FEC	10.6	RMSE (W)	16.25	13.03	11.39	15.01	18.93	11.45
			MAE (W)	12.35	10.99	9.14	13.81	15.17	8.45
			R ² (%)	86.22	87.95	89.37	85.47	82.09	91.27
			PINAW	0.34	0.33	0.22	0.36	0.33	0.22
			CRPS	16.56	14.24	10.95	19.41	15.76	8.63
	M-FEC	22.3	RMSE (W)	27.66	23.19	22.22	26.28	25.27	14.95
			MAE (W)	21.93	15.93	15.41	20.03	22.96	13.37
			R ² (%)	77.74	80.90	81.55	76.66	74.38	89.34
			PINAW	0.51	0.54	0.48	0.55	0.53	0.31
			CRPS	26.77	23.43	23.16	24.38	25.78	10.23
	I-FEC	38.8	RMSE (W)	35.01	35.97	33.43	39.76	43.76	22.76
			MAE (W)	30.07	26.83	25.25	29.55	33.13	20.62
			R ² (%)	69.03	72.14	73.12	67.00	66.51	81.50
			PINAW	0.69	0.57	0.56	0.67	0.61	0.39
			CRPS	31.32	29.92	30.20	30.13	33.82	13.75

reformulates them into a quadratic form, thereby enabling accurate capture of abrupt fluctuations. Meanwhile, the FAM quantifies fluctuation intensity using a fluctuation sensitivity coefficient λ and adaptively routes fluctuations of different intensities to dedicated prediction branches. In this way, the model specifically learns fluctuation features under S-FEC, M-FEC, and I-FEC, while effectively representing stochastic fluctuations. In the experimental evaluation, PKINN was compared with five SOTA forecasting models on two PV modules. The results demonstrate that PKINN consistently delivers superior performance across all fluctuation intensities, achieving an average RMSE of 8.42 W, MAE of 6.50 W, and R^2 of 91.62%. Notably, under I-FEC, PKINN achieves more than a 13.6% reduction in RMSE relative to the best-performing comparative model. These findings highlight the framework's advantages in forecasting accuracy and interpretability under FECs.

CRediT authorship contribution statement

Jialong Pei: Writing – original draft, Software, Methodology, Conceptualization. **Jieming Ma:** Writing – review & editing, Visualization, Supervision, Conceptualization. **Ka Lok Man:** Writing – review

& editing, Validation, Supervision. **Martin Gairing:** Writing – review & editing, Validation, Supervision.

Code and data availability

The source code developed and used in this study is openly accessible at the following GitHub repository: https://github.com/addm9/Photovoltaic_Knowledge_Informed_Neural_Network.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table A.1

Definitions of symbols, constants, and their SI units.

Symbol	Description	Unit (SI)
I_{ph}	Photocurrent	A
I_{sat}	Saturation current	A
I_{ref}	Reference current obtained from QEM	A
I_{emp}	Empirical current from SDM	A
V	Terminal voltage	V
R_s	Series resistance	Ω
R_{sh}	Shunt resistance	Ω
A	Diode ideality factor	–
$V_t = kT/q$	Thermal voltage	V
k	Boltzmann constant (1.381×10^{-23} J/K)	–
q	Electron charge (1.602×10^{-19} C)	–
T	Cell temperature	K
I_0	Saturation current at reference conditions	A
I_{sc}	Short-circuit current	A
V_{oc}	Open-circuit voltage	V

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Appendix. Derivation and verification of the Quadratic Explicit Model (QEM)

This appendix has been newly added to provide the detailed derivation and validation of the QEM. The QEM reformulates the nonlinear single-diode model into an explicit quadratic algebraic form to describe the intrinsic I–V characteristics of PV modules under FECs.

A.1. Definition of symbols, constants, and units

Table A.1 summarizes the symbols, physical constants, and variables used in the derivation of the Quadratic Explicit Model (QEM), together with their corresponding SI units.

A.2. Derivation of the quadratic explicit form

The QEM is constructed from the fundamental single-diode model, which characterizes the I–V relationship of PV modules under varying environmental conditions. Its form is expressed as:

$$I = I_{ph} - I_{sat} \left[\exp \left(\frac{V + IR_s}{AV_t} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}}. \quad (A.1)$$

where all symbols are defined in Table A.1. This nonlinear transcendental equation cannot be solved explicitly for I because of the exponential term that couples I and V . Differentiating (A.1) with respect to V gives the voltage derivative of current I :

$$\frac{\partial I}{\partial V} = - \frac{I_{sat} R_s R_{sh} \exp \left(\frac{V + IR_s}{AV_t} \right) + AV_t}{I_{sat} R_s R_{sh} \exp \left(\frac{V + IR_s}{AV_t} \right) + AV_t (R_s + R_{sh})}. \quad (A.2)$$

Since the instantaneous power is $P = VI$, the variation of P with respect to voltage reflects the condition for reaching the maximum power point, and the power derivative can be written as $\frac{\partial P}{\partial V} = V \frac{\partial I}{\partial V} + I$. Replacing $\frac{\partial I}{\partial V}$ in this expression with (A.2) yields:

$$\frac{\partial P}{\partial V} = I - V \frac{I_{sat} R_s R_{sh} \exp \left(\frac{V + IR_s}{AV_t} \right) + AV_t}{I_{sat} R_s R_{sh} \exp \left(\frac{V + IR_s}{AV_t} \right) + AV_t (R_s + R_{sh})}. \quad (A.3)$$

At the maximum power point, $\frac{\partial P}{\partial V} = 0$, indicating that the operating condition corresponds to the extremum of the P – V curve. Under this

optimality condition, (A.3) can be rearranged to give an analytically equivalent expression for the exponential term as:

$$\exp \left(\frac{V + IR_s}{AV_t} \right) = \frac{AV_t (V - (R_s + R_{sh})I)}{I_{sat} R_{sh} (R_s I - V)}. \quad (A.4)$$

By substituting (A.4) into (A.1) and rearranging the terms, a quadratic equation with respect to the reference current I_{ref} can be derived as $k_0 I_{ref}^2 + k_1 I_{ref} + k_2 = 0$, where the coefficients are defined as:

$$\begin{cases} k_0 = -R_s(R_s + R_{sh}), \\ k_1 = (I_{sc} + I_0)R_s R_{sh} + (AV_t + V)(R_s + R_{sh}) - R_s V, \\ k_2 = V^2 - [(I_0 + I_{sc})R_s R_{sh} + AV_t]V. \end{cases} \quad (A.5)$$

By applying the quadratic formula, the explicit expression of the reference current is obtained:

$$I_{ref} = \frac{1}{2k_0} \left(-k_1 + \sqrt{k_1^2 - 4k_0 k_2} \right). \quad (A.6)$$

This expression defines the reference current I_{ref} . Through this formulation, the nonlinear single-diode model is transformed into an analytically solvable quadratic algebraic form. Based on I_{ref} , the QEM further constructs the current deviation function $D_{cur}(V) = |I_{emp}(V) - I_{ref}(V)|$, which serves as the optimization criterion for identifying the theoretical target point on the P – V curve and for guiding the iterative refinement process within the model.

A.3. Dimensional and numerical consistency verification

To ensure the overall consistency of the QEM formulation, its analytical structure and numerical behavior were examined through both dimensional and numerical analyses.

(a) Dimensional verification. In the quadratic expression of the QEM, $k_0 I_{ref}^2 + k_1 I_{ref} + k_2 = 0$, all terms must possess the same physical unit to ensure dimensional consistency. By dimensional expansion:

$$\begin{aligned} [k_0 I_{ref}^2] &= [\Omega^2 A^2] = [V A^{-1}], & [k_1 I_{ref}] &= [\Omega V A^{-1} \cdot A] = [V A^{-1}], \\ [k_2] &= [V^2 A^{-1}] = [V A^{-1}]. \end{aligned}$$

Since all three terms yield the same unit of $[V A^{-1}]$, the equation is dimensionally homogeneous, confirming the internal unit consistency of the QEM.

(b) Numerical cross-validation. To further assess the analytical reliability, the QEM was validated using experimental I–V data collected from multiple PV modules. The power values computed from Eq. (A.6) were compared with the measured power, showing an average deviation of less than 0.8% across the effective voltage range. This slight deviation demonstrates the numerical stability of the QEM and indicates that the quadratic approximation can accurately reproduce the operating behavior of the PV modules.

Data availability

The source code developed and used in this study is openly accessible at the following GitHub repository:

https://github.com/addm9/Photovoltaic_Knowledge_Informed_Neural_Network.

References

- [1] Kallio S, Siroux M. Photovoltaic power prediction for solar micro-grid optimal control. *Energy Rep* 2023;9:594–601. <http://dx.doi.org/10.1016/j.egy.2022.11.081>.
- [2] Liu H, Gao Q, Ma P. Photovoltaic generation power prediction research based on high quality context ontology and gated recurrent neural network. *Sustain Energy Technol Assessments* 2021;45:101191. <http://dx.doi.org/10.1016/j.seta.2021.101191>.

- [3] Xu X, Chen Y, Goude Y, Brossat X, Antoniadis A. Day-ahead probabilistic forecasting for french half-hourly electricity loads and quantiles for curve-to-curve regression. *Appl Energy* 2021;301:117465. <http://dx.doi.org/10.1016/j.apenergy.2021.117465>.
- [4] Paraschiv S, Paraschiv LS, Serban A. Increasing the energy efficiency of a building by thermal insulation to reduce the thermal load of the micro-combined cooling, heating and power system. *Energy Rep* 2021;7:286–98. <http://dx.doi.org/10.1016/j.egy.2021.07.122>.
- [5] Sun F, Li L, Bian D, et al. Photovoltaic power prediction based on multi-scale photovoltaic power fluctuation characteristics and multi-channel LSTM prediction models. *Renew Energy* 2025;246:122866. <http://dx.doi.org/10.1016/j.renene.2025.122866>.
- [6] Tripathi AK, Aruna M, Elumalai PV, et al. Advancing solar PV panel power prediction: A comparative machine learning approach in fluctuating environmental conditions. *Case Stud Therm Eng* 2024;59:104459. <http://dx.doi.org/10.1016/j.csite.2024.104459>.
- [7] Zhu J, Li M, Luo L, et al. Short-term PV power forecast methodology based on multi-scale fluctuation characteristics extraction. *Renew Energy* 2023;208:141–51. <http://dx.doi.org/10.1016/j.renene.2023.03.029>.
- [8] Hassan MA, Bailek N, Bouchouicha K, et al. Ultra-short-term exogenous forecasting of photovoltaic power production using genetically optimized non-linear auto-regressive recurrent neural networks. *Renew Energy* 2021;171:191–209. <http://dx.doi.org/10.1016/j.renene.2021.02.103>.
- [9] Zhen H, Niu D, Wang K, et al. Photovoltaic power forecasting based on GA improved bi-LSTM in microgrid without meteorological information. *Energy* 2021;231:120908. <http://dx.doi.org/10.1016/j.energy.2021.120908>.
- [10] Batzels EI, Kampitsis GE, Papathanassiou SA, et al. Direct MPP calculation in terms of the single-diode PV model parameters. *IEEE Trans Energy Convers* 2015;30(1):226–36. <http://dx.doi.org/10.1109/TEC.2014.2356017>.
- [11] Salah FEA, Maouhoub N, Tifidat K, et al. An efficient analytical approach for forecasting the peak power of PV panels working in outdoor conditions based on explicit model. *Energy Convers Manag* 2023;20:100423. <http://dx.doi.org/10.1016/j.ecmx.2023.100423>.
- [12] Pereira S, Canhoto P, Salgado R. Development and assessment of artificial neural network models for direct normal solar irradiance forecasting using operational numerical weather prediction data. *Energy AI* 2024;15:100314. <http://dx.doi.org/10.1016/j.egyai.2024.100314>.
- [13] Lopes BM, Costa AMG, Uturbey W. Simulation with performance evaluation of photovoltaic production using long term monthly irradiation data. *Renew Energy* 2024;223:120021. <http://dx.doi.org/10.1016/j.renene.2024.120021>.
- [14] Mayer MJ, Yang D. Pairing ensemble numerical weather prediction with ensemble physical model chain for probabilistic photovoltaic power forecasting. *Renew Sustain Energy Rev* 2023;175:113171. <http://dx.doi.org/10.1016/j.rser.2023.113171>.
- [15] Zhi Y, Sun T, Yang X. A physical model with meteorological forecasting for hourly rooftop photovoltaic power prediction. *J Build Eng* 2023;75:106997. <http://dx.doi.org/10.1016/j.job.2023.106997>.
- [16] Koster D, Minette F, Braun C, et al. Short-term and regionalized photovoltaic power forecasting, enhanced by reference systems, on the example of Luxembourg. *Renew Energy* 2019;132:455–70. <http://dx.doi.org/10.1016/j.renene.2018.08.005>.
- [17] Ma J, Wang K, Huang M, et al. V cp-decomposition-based adaptive search for optimal power extraction in photovoltaic systems. *IEEE Trans Power Electron* 2024;39(10):12794–802. <http://dx.doi.org/10.1109/TPEL.2024.3423485>.
- [18] Lauret P, Perez R, David M, et al. The added value of combining solar irradiance data and forecasts: A probabilistic benchmarking exercise. *Renew Energy* 2024;237:121574. <http://dx.doi.org/10.1016/j.renene.2023.121574>.
- [19] Fan GF, Han YY, Li JW, et al. A hybrid model for deep learning short-term power load forecasting based on feature extraction statistics techniques. *Expert Syst Appl* 2024;238:122012. <http://dx.doi.org/10.1016/j.eswa.2023.122012>.
- [20] Pan M, Li W, Chen H, et al. Photovoltaic power forecasting based on a support vector machine with improved ant colony optimization. *J Clean Prod* 2020;277:123948. <http://dx.doi.org/10.1016/j.jclepro.2020.123948>.
- [21] Chen X, Ding K, Zhang J, et al. Online prediction of ultra-short-term photovoltaic power using chaotic characteristic analysis, improved PSO and KELM. *Energy* 2022;248:123574. <http://dx.doi.org/10.1016/j.energy.2022.123574>.
- [22] Majumder I, Dash PK, Bisoi R. Variational mode decomposition based low rank robust kernel extreme learning machine for solar irradiation forecasting. *Energy Convers Manage* 2018;171:787–806. <http://dx.doi.org/10.1016/j.enconman.2018.06.021>.
- [23] Li N, Li L, Zhang F, et al. Research on short-term photovoltaic power prediction based on multi-scale similar days and ESN-KELM dual core prediction model. *Energy* 2023;277:127557. <http://dx.doi.org/10.1016/j.energy.2023.127557>.
- [24] Luo X, Zhang D, Zhu X. Deep learning based forecasting of photovoltaic power generation by incorporating domain knowledge. *Energy* 2021;225:120240. <http://dx.doi.org/10.1016/j.energy.2021.120240>.
- [25] Ge L, Xian Y, Yan J, et al. A hybrid model for short-term PV output forecasting based on PCA-gwo-GRNN. *J Mod Power Syst Clean Energy* 2020;8(6):1268–75. <http://dx.doi.org/10.35833/MPCE.2020.000004>.
- [26] Yuan L, Wang X, Sun Y, et al. Multistep photovoltaic power forecasting based on multi-timescale fluctuation aggregation attention mechanism and contrastive learning. *Int J Electr Power Energy Syst* 2025;164:110389. <http://dx.doi.org/10.1016/j.ijepes.2024.110389>.
- [27] Chen Z, Bai Y, Hong J. Constructing two-stream input matrices in a convolutional neural network for photovoltaic power prediction. *Eng Appl Artif Intell* 2024;135:108814. <http://dx.doi.org/10.1016/j.engappai.2024.108814>.
- [28] Walters M, Venayagamoorthy GK. Situational awareness indices of solar PV power generation under temporal weather conditions for near real-time planning and operation. *Appl Energy* 2025;402:126855. <http://dx.doi.org/10.1016/j.apenergy.2024.126855>.
- [29] Tan L, Kang R, Xia J, et al. Application of multi-source data fusion on intelligent prediction of photovoltaic power. *Sol Energy* 2024;277:112706. <http://dx.doi.org/10.1016/j.solener.2023.112706>.
- [30] Zhang J, Liu Z, Chen T. Interval prediction of ultra-short-term photovoltaic power based on a hybrid model. *Electr Power Syst Res* 2023;216:109035. <http://dx.doi.org/10.1016/j.epsr.2022.109035>.
- [31] Hong D, Ma J, Wang K, et al. Real-time power prediction for bifacial PV systems in varied shading conditions: A circuit-LSTM approach within a digital twin framework. *IEEE J Photovoltaics* 2024;14(4):652–60. <http://dx.doi.org/10.1109/JPHOTOV.2024.3393001>.
- [32] Wang G, Sun S, Fan S, et al. A spatial-temporal data-driven deep learning framework for enhancing ultra-short-term prediction of distributed photovoltaic power generation. *Int J Electr Power Energy Syst* 2024;160:110125. <http://dx.doi.org/10.1016/j.ijepes.2023.110125>.
- [33] Song Z, Xiao F, Chen Z, et al. Probabilistic ultra-short-term solar photovoltaic power forecasting using natural gradient boosting with attention-enhanced neural networks. *Energy AI* 2025;20:100496. <http://dx.doi.org/10.1016/j.egyai.2024.100496>.
- [34] Yong B, Zhang Y, Shen J, et al. Convode-mixer: A multimodal deep learning model for ultra-short-term PV power forecasting. *Sol Energy* 2025;300:113777. <http://dx.doi.org/10.1016/j.solener.2024.113777>.
- [35] Abuella M, Chowdhury B. Forecasting of solar power ramp events: A post-processing approach. *Renew Energy* 2019;133:1380–92. <http://dx.doi.org/10.1016/j.renene.2018.09.005>.
- [36] Raissi M, Perdikaris P, Karniadakis GE. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J Comput Phys* 2019;378:686–707. <http://dx.doi.org/10.1016/j.jcp.2018.10.045>.
- [37] Cheng S, Alkhalifah T. Meta-PINN: Meta learning for improved neural network wavefield solutions. 2024, arXiv preprint, [arXiv:2401.11502](https://arxiv.org/abs/2401.11502).
- [38] Zhang X, Mao B, Che Y, et al. Physics-informed neural networks (PINNs) for 4D hemodynamics prediction: An investigation of optimal framework based on vascular morphology. *Comput Biol Med* 2023;164:107287. <http://dx.doi.org/10.1016/j.combiomed.2023.107287>.
- [39] Li S, Li X, Jiang Y, et al. A novel frequency-domain physics-informed neural network for accurate prediction of 3D spatio-temporal wind fields in wind turbine applications. *Appl Energy* 2025;386:125526. <http://dx.doi.org/10.1016/j.apenergy.2024.125526>.
- [40] Chen L, Wang Y, Sun W, et al. Modified hard-constrained PINNs for physically consistent microgrid transient response modeling. *Int J Electr Power Energy Syst* 2025;169:110726. <http://dx.doi.org/10.1016/j.ijepes.2024.110726>.
- [41] Sharadga H, Hajimirza S, Balog RS. Time series forecasting of solar power generation for large-scale photovoltaic plants. *Renew Energy* 2020;150:797–807. <http://dx.doi.org/10.1016/j.renene.2019.12.131>.
- [42] Huang B, Wang J. Applications of physics-informed neural networks in power systems—a review. *IEEE Trans Power Syst* 2022;38(1):572–88. <http://dx.doi.org/10.1109/TPWRS.2022.3162473>.
- [43] Ma J, Hong D, Wang K, et al. Analytical modeling and parameter estimation of photovoltaic strings under partial shading conditions. *Sol Energy Mater Sol Cells* 2022;235:111494. <http://dx.doi.org/10.1016/j.solmat.2021.111494>.
- [44] Sarvi M, Azadian A. A comprehensive review and classified comparison of MPPT algorithms in PV systems. *Energy Syst* 2022;13(2):281–320. <http://dx.doi.org/10.1007/s12667-021-00427-x>.
- [45] Lappalainen K, Wang GC, Kleissl J. Estimation of the largest expected photovoltaic power ramp rates. *Appl Energy* 2020;278:115636. <http://dx.doi.org/10.1016/j.apenergy.2020.115636>.
- [46] Liao Z, Coimbra CFM. Hybrid solar irradiance nowcasting and forecasting with the SCOPE method and convolutional neural networks. *Renew Energy* 2024;232:121055. <http://dx.doi.org/10.1016/j.renene.2023.121055>.
- [47] Djeldjeli Y, Khezami N, Bouchouicha K, et al. Enhancing solar power forecasting with machine learning using principal component analysis and diverse statistical indicators. *Case Stud Therm Eng* 2024;61:104924. <http://dx.doi.org/10.1016/j.csite.2024.104924>.
- [48] Wan A, et al. Short-term power load forecasting for combined heat and power using CNN-LSTM enhanced by attention mechanism. *Energy* 2023;282:128274. <http://dx.doi.org/10.1016/j.energy.2023.128274>.
- [49] Lu H, et al. Evolution of the shape parameters of photovoltaic module as a function of temperature and irradiance: New method of performance prediction without setting reference conditions. *Energy Rep* 2022;8:13859–75. <http://dx.doi.org/10.1016/j.egy.2022.10.012>.

- [50] Li J, et al. Prediction of I–V characteristic curve for photovoltaic modules based on convolutional neural network. *Sensors* 2020;20(7):2119. <http://dx.doi.org/10.3390/s20072119>.
- [51] Verdone A, et al. A review of solar and wind energy forecasting: From single-site to multi-site paradigm. *Appl Energy* 2025;392:126016. <http://dx.doi.org/10.1016/j.apenergy.2024.126016>.
- [52] Ghimire S, et al. Point-based and probabilistic electricity demand prediction with a neural facebook prophet and kernel density estimation model. *Eng Appl Artif Intell* 2024;135:108702. <http://dx.doi.org/10.1016/j.engappai.2024.108702>.
- [53] Du J, Zheng J, Liang Y, et al. A theory-guided deep-learning method for predicting power generation of multi-region photovoltaic plants. *Eng Appl Artif Intell* 2023;118:105647. <http://dx.doi.org/10.1016/j.engappai.2022.105647>.
- [54] Yu S, He B, Fang L. Multi-step short-term forecasting of photovoltaic power utilizing TimesNet with enhanced feature extraction and a novel loss function. *Appl Energy* 2025;388:125645. <http://dx.doi.org/10.1016/j.apenergy.2024.125645>.
- [55] Lin S, Lin W, Wu W, et al. Segrnn: Segment recurrent neural network for long-term time series forecasting. 2023, arXiv preprint, [arXiv:2308.11200](https://arxiv.org/abs/2308.11200).
- [56] Peng T, Fu Y, Wang Y, et al. An intelligent hybrid approach for photovoltaic power forecasting using enhanced chaos game optimization algorithm and locality sensitive hashing based informer model. *J Build Eng* 2023;78:107635. <http://dx.doi.org/10.1016/j.jobbe.2023.107635>.
- [57] Piantadosi G, Dutto S, Galli A, et al. Photovoltaic power forecasting: A transformer based framework. *Energy AI* 2024;100444. <http://dx.doi.org/10.1016/j.egyai.2024.100444>.